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A palaeomagnetic and palaeointensity study of the Mauchline lavas, south-west Scotland

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Abstract

The Mauchline lavas of SW Scotland erupted close to the Carboniferous–Permian boundary and near the middle of the Permo–Carboniferous Reversed Superchron (PCRS). Rock magnetic studies have identified mixed magnetite–hematite assemblages; both minerals record reversed components of magnetisation. A fold test is not diagnostic but the mean remanence recovered in 23 stable samples at five sites ($D/I = 192.5/-26.1°$, $\alpha_{95} = 14.3°$) accords with the regional field at the time of eruption and the pole position (346.8°E, 47.2°S) is coincident with late Carboniferous–Early Permian poles. A palaeointensity study of 15 samples employing the modified Thellier–Thellier method yields nine acceptable estimates of ancient field intensity with a mean value of 13.6 ± 4.5 μT from four lava flows. The corresponding virtual dipole moment of 3.3×10^{22} Am^2 is approximately 40% of the present day value. This result supports recent studies from western Europe which suggest that the geomagnetic field strength was low during the PCRS but is in conflict with Russian data which suggest that the field strength was close to, or slightly higher than the present day value.

1. Introduction

The most important temporal feature of geomagnetic field polarity is the alternation of periods of frequent inversion with protracted periods when no inversions appear to have taken place. Two long periods of constant geomagnetic field polarity (superchrons) are recognised during the Phanerozoic eon (Piper, 1987). The most recent, the Cretaceous Normal Superchron (CNS), occupied the interval between Aptian and Santonian times (ca. 118 and 83 Ma) and it is well constrained from the ocean floor magnetic anomaly signature as well as from numerous palaeomagnetic studies on land. The second is the Permo–Carboniferous Reversed Superchron (PCRS). This was recognised by Graham (1955) in Permian rocks of North America and subsequently constrained by Irving and Parry (1963) to the interval between Lower Carboniferous and Upper Permian times. Study of the late Permian Dewey Lake Formation of north-west Texas recognised a termination of this superchron at about 251 ± 4 Ma near the boundary between Lower and Upper Tartarian times (Molina-Garza et al., 1989). The lower boundary is less well

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constrained but is probably located within late Westphalian 'A' or early Westphalian 'B' times (DiVenere and Opdyke, 1991) in early Upper Carboniferous times at ca. 320 Ma (Harland et al., 1989). These long periods of constant geomagnetic polarity are both followed by periods showing a linear increase in reversal rate. Analysis of this variable reversal rate identifies an underlying time period of 108 years similar to the overturn time of mantle convection (McFadden and Merrill, 1986; Piper, 1987). The link between reversal rate and mantle convection is widely related to the D″ layer just above the core–mantle boundary (CMB) because variations in thickness of this layer will clearly influence core–mantle heat flow. Currently two models have been invoked to explain this link. Larson and Olson (1991) suggest that increased heat flow across a thin core–mantle boundary creates more vigorous convection in the outer core. The associated heat loss across the CMB would lead to a strong, stable, single-polarity field. Conversely, a thick CMB would minimise heat loss and reduce core convection, leading to increased reversal frequency. In contrast, Loper and McCartney (1986) argue that increased convection in the outer core would increase reversal frequency by generating instabilities. The Larson and Olson model predicts an inverse relationship between field strength and reversal frequency. Palaeointensity studies of rocks formed during the Cretaceous Normal Superchron (Prévot et al., 1990; Pick and Tauxe, 1994) do not support this model and suggest that the processes controlling field strength and reversal frequency are, in fact, decoupled. We report a palaeomagnetic and palaeointensity study of lavas from Scotland erupted during the PCRS and compare the data with other palaeointensity results from this superchron in order to add to our knowledge of the field strength during this long period of constant polarity.

2. Geology and sampling

At Mauchline in the south-western part of the Midland Valley of Scotland (355.4°E, 55.5°N, see Fig. 1) a shallow NW–SE trending basin of late Carboniferous–Permian rocks lies unconformably on Coal Measures of Upper Carboniferous age. A succession of basaltic lavas increases in thickness from 100 to 240 m from west to east and is overlain by red aeolian and water-lain sandstones (Eyles et al., 1949). Numerous small volcanic necks in a 20 km radius around the basin define the former extent of a volcanic field. The lavas are strongly undersaturated olivine basalts and typically heavily weathered with a thick soil cover; as a result fresh in situ exposure is scarce and only five outcrops (Fig. 1 and Table 1) proved suitable for palaeomagnetic investigation. Where unweathered material is recovered, olivine is the only primary silicate appreciably altered (Eyles et al., 1949).

The presence of wind blown sands both between and above the lavas indicates a dry arid

Table 1
Site location data with site and group mean palaeomagnetic results following thermal cleaning

Site No.	Grid.	Ref.	No. cores	N/n	R	k	α_{95}	D (in situ)	I (in situ)	D (tilt adj.)	I (tilt adj.)
1	2515	6255	17	6/5	4.95	85	8	187.5	−40.9	179.9	−44.6
2	2433	6275	17	6/2	1.96	25	–	198.3	−31.3	200.4	−22.4
3	2477	6210	19	6/5	4.86	28	15	204.1	−12.2	203.5	−5.3
4	2510	6275	20	6/6	5.64	14	19	194.3	−27.9	191.4	−34.4
5	2445	6255	12	6/5	4.86	29	14	177.6	−16.8	179.7	−12.5
Group mean result, 5 sites:											
In situ:					4.97	29.7	14.3	192.5	−26.1		
(Palaeomagnetic pole: 346.8°E, 47.2°S, dp/dm = 8.3/15.3°)											
Tilt adjusted:					4.78	18.5	18.2	191.6	−24.1		
(Palaeomagnetic pole: 348.0°E, 46.2°S, dp/dm = 10.5/19.7°)											

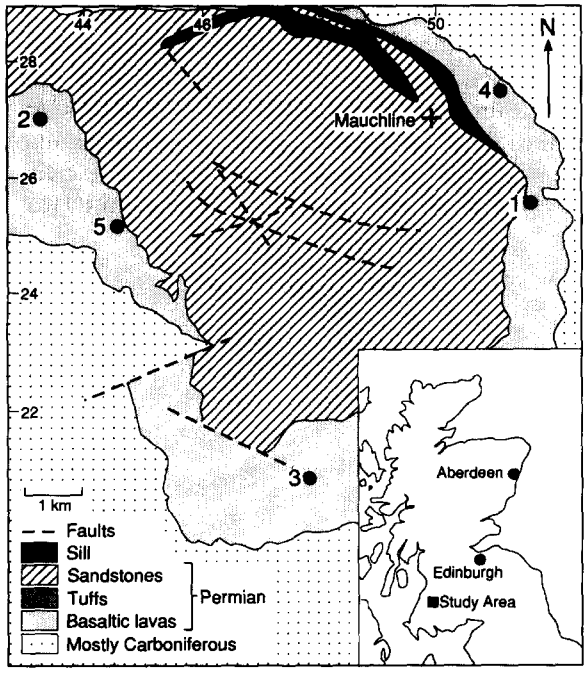


Fig. 1. Outline geological map of the Mauchline region, SW Scotland, based on Eyles et al. (1949) and showing the location of the sampling sites for this study. The inset map shows the regional location.

environment and implies a correlation with small late Carboniferous and early Permian basins of aeolian sands elsewhere in Britain; hence these rocks appear to predate the Upper Permian–Lower Triassic Bunter and Keuper successions (Eyles et al., 1949; Pattison et al., 1973). Although the Carboniferous–Permian boundary has conventionally been placed at the contact between the lavas and the underlying Coal Measures, the presence of a Stephanian plant flora between flow units (Craig, 1970) implies eruption late in Carboniferous times (ca. 290 Ma, Harland et al., 1989) approximately 20 Ma after commencement of the PCRS.

Du Bois (1957) reported total natural remanent magnetisation (NRM) directions from 34 cores taken from unlocated block samples of these lavas; all directions were southerly shallow negative (mean $D/I = 180/-4°$, $\alpha_{95} = 8°$) close to the late Palaeozoic reversed field direction for Britain. Hence it is likely that remanence is dominated by components with an age close to the time of eruption; they are therefore suitable targets for palaeointensity investigation. Samples for the present study were drilled by portable drilling motor; between 12 and 20 separate 2.4-cm diameter cores were distributed over available outcrops and oriented in situ using sun and magnetic compasses. Site locations are listed as grid references in Table 1 and shown in Fig. 1. The collection was divided into core sets for palaeomagnetic and palaeointensity experiments.

3. Palaeomagnetic results

3.1. Magnetic mineralogy

Rock magnetic investigations are an essential complement to palaeointensity studies. In gen-

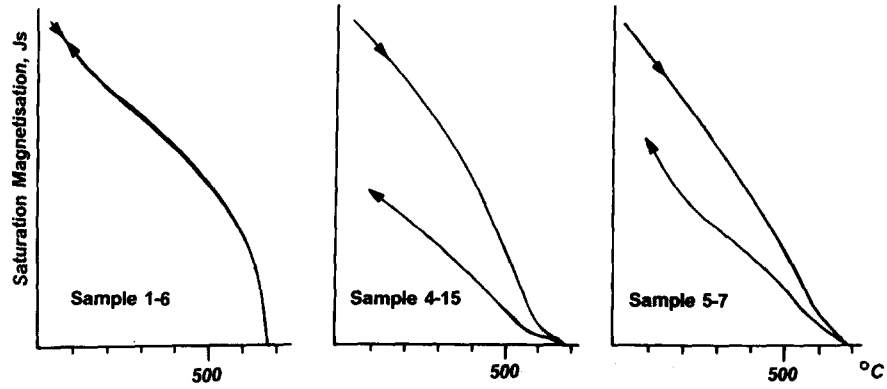


Fig. 2. Thermomagnetic determinations (saturation magnetisation, *Js*, versus temperature) illustrating characteristic curves derived from the Mauchline lavas.

eral, they are required to isolate material which is able to survive successive heating experiments without major chemical change. Specifically, they must identify the nature (composition, domain states and oxidation states) of the remanence carriers. Whilst only single domain (SD) carriers are able to preserve a reliable record of the ancient field strength, Thomas (1993) finds that mixed multidomain (MD):single domain assemblages yield reproducible results provided that significant proportions of SD carriers are present. If the Shaw (alternating field – a.f.) technique (Shaw, 1974) is to be applied a broad coercivity spectrum must also be present in magnetite (Ro-

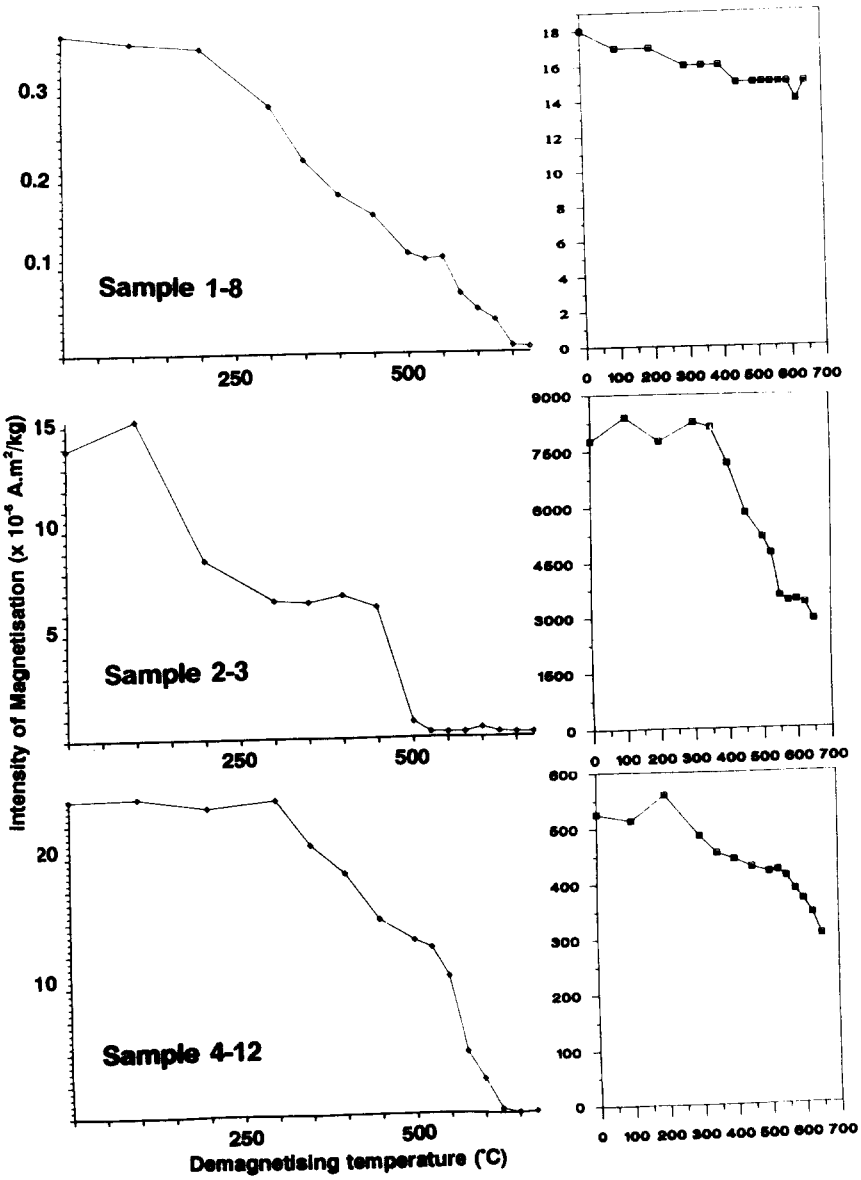


Fig. 3. Representative examples of reduction in intensity of magnetisation, J ($Am^2\ kg^{-1}$), with progressive thermal demagnetisation and associated changes in susceptibility (S.I. units).

berts and Shaw, 1990). This will not be the case when hematite is a significant contributor to the magnetic mineral assemblage since conventional a.f. demagnetisers are unable to supply fields large enough to completely demagnetise hematite.

Thermomagnetic determinations (saturation magnetisation, *Js*, versus temperature, *T*) were conducted on millimetre size fragments from 16 cores. Following the classification of Mankinen et al. (1985) two behaviours, 2a and 2c, are evident (Fig. 2). Type 2a illustrates near-identical curves for heating and cooling of the sample with the final magnetisation, $Js_f > 10\%$ lower than the initial magnetisation, Js_i and is observed at Site 1. In samples from the remaining sites Type 2c behaviour is defined by a marked change in shape of the cooling curve with $Js_f > 10\%$ lower than Js_i. Strong saturation remanences dominated by magnetite are appreciably lowered on cooling probably due to partial conversion to hematite which has a much lower saturation remanence. In some examples significant hematite is identified on heating as a tail to the curve; in others this is barely perceptible (Fig. 2).

Hence hematite is a significant magnetic constituent of these lavas. Its presence is confirmed by distributed unblocking temperature spectra extending above the Curie point of magnetite (580°C) in all lavas except Site 2 (Fig. 3) which has lower unblocking temperature spectra with remanence fully unblocked by 520°C. Samples from Site 2 undergo prominent chemical change with progressive thermal treatment identified by a sharp susceptibility drop above 350°C (Fig. 3). The remaining sites show no change (Site 1) or small progressive reduction in susceptibility (Sites 3–5) at this temperature and above. Since all remanence is subtracted by 650°C (Fig. 4), the hematite is evidently titaniferous.

Thus rock magnetic experiments identify the need for thermal demagnetisation to isolate component structure and show that only the Thellier technique (Thellier and Thellier, 1959) is applicable to palaeointensity investigation of these rocks. The broad size range of SD behaviour in hematite however (ca. 0.02 to > 15 μm), is promising for palaeointensity study provided that the remanence can be shown to have recorded a primary palaeofield.

3.2. Palaeodirectional study

Palaeomagnetic analysis employed thermal demagnetisation steps of 100°C up to 300°C and then 50 or 20°C steps to the Curie points of the

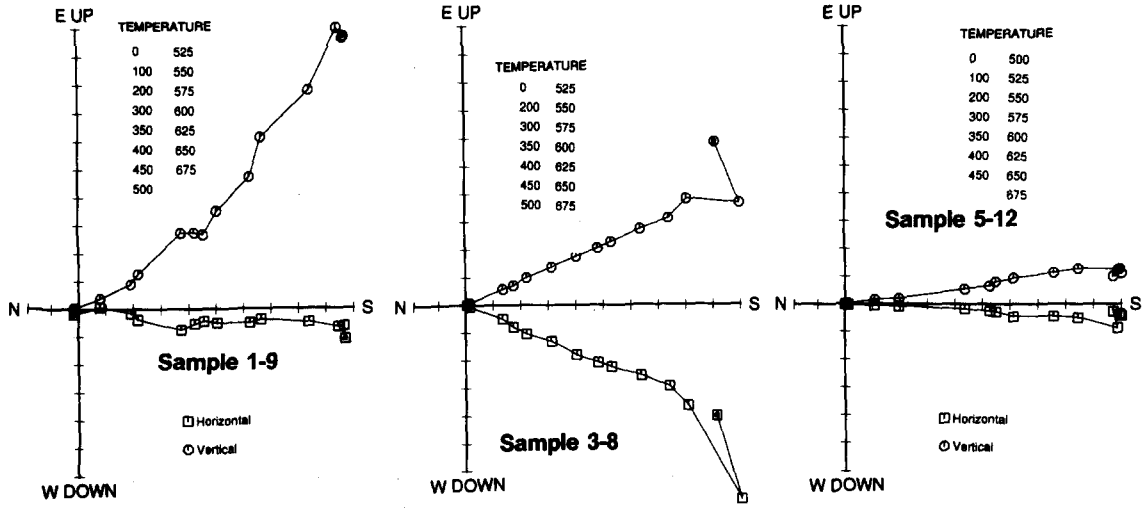


Fig. 4. Orthogonal vector plots illustrating progressive thermal demagnetisation of samples from the Mauchline lavas. Squares are projections onto the horizontal plane and circles are projections onto the vertical plane.

remanence carriers. Possible chemical change during demagnetisation was monitored at each step by susceptibility measurements.

Progressive thermal demagnetisation identifies two or three components in all samples (Fig. 4). Lower blocking temperature (lbt) components are small and subtracted by 300°C. Higher blocking temperature (hbt) components are resident in magnetite and hematite at Sites 1 and 3–5. At the latter three sites magnetite and hematite-held remanences have the same directions, suggesting that high-temperature deuteric oxidation has occurred. At Site 1 the magnetite remanence is of steeper inclination (Sample 1–9 in Fig. 4 and Table 1). At Site 2 magnetite-held remanence was poorly stable; vectors were noisy with only two samples yielding significant components ($\alpha_{95} < 20°$, Table 1). Directions of the hbt components are all southerly and most have shallow negative inclinations (Fig. 5).

Site mean directions (Table 1) yield a group mean direction of $D/I = 192.5/-26.1°$. The difference between this result and the total NRM result ($D/I = 180/-4°$) of Du Bois (1957) is accounted for by inclusion of lbt (and positive inclination) components of more recent origin in the total NRM result. Dispersion of site means in situ is somewhat larger than expected from the effects of secular variation at a palaeolatitude of 13.5° and suggests that tilt adjustment may be applicable here. Unfortunately the paucity of outcrop does not allow this to be confidently as-

Table 2
Palaeomagnetic results from Carboniferous–Permian igneous rocks of the British Isles derived from thermal demagnetisation and component analysis

Rock unit	Pole position		Age (Ma)	Ref.	Remanence (primary/sec)
	°E	°S			
(i) Lower Carboniferous					
1. Derbyshire lavas	336	14	ca. 335	1	*
2. Derbyshire intrusions	329	12		1	*
3. Kinghorn Lavas	343	18		2	*
4. Kinghorn-B'land Lavas	332	14		3	*
5. Largs Lavas	322	14		3	*
6. Big Dyke, Dunbar	341	23		4	*
(ii) Upper Carboniferous-Permian					
7. Dykes, NW Highlands	355	35		5	*
8. Derbyshire SE intrusions	304	38	309	1	*
9. Derbyshire overprint	313	42		1	* *
10. Volcanics, Limerick	329	39		6 [a]	* *
11. Sill, Queensferry	354	38		3	*
12. Dyke, Peterhead	342	41		7	*
13. Overprints, Largs	343	43		3	* *
14. Whin Sill	339	44		8	*
15. Exeter Traps	345	46	281,279	9	*
16. Mauchline Lavas	347	47		10	*
17. Dykes, Outer Hebrides	347	47		11	*
18. Wackerfield dyke	349	49	303	12	*
19. Small dyke, Dunbar	344	51		4	* *
20. Midland Valley Sill	351	41		14	*
21. Great Whin Sill	348	37	301	14	*
22. Great Whin Sill	353	37		13	*

[a] A.f. demagnetisation.

References to the palaeomagnetic studies are: 1, Piper et al. (1991); 2, Wilson and Everitt (1963); 3, Torsvik et al. (1989); 4. Xu and Tarling (1987); 5, Esang and Piper (1984); 6, Muller (1972); 7, Torsvik (1985); 8, Storetvedt and Gidskehaug (1969); 9, Cornwell (1967) and Zijderveld (1967); 10, This study; 11, Piper (1992); 12, Tarling and Mitchell (1973); 13, Thomas et al. (1995a); 14, Creer et al. (1959). Remanence is interpreted by the authors as primary (*) or secondary (**).

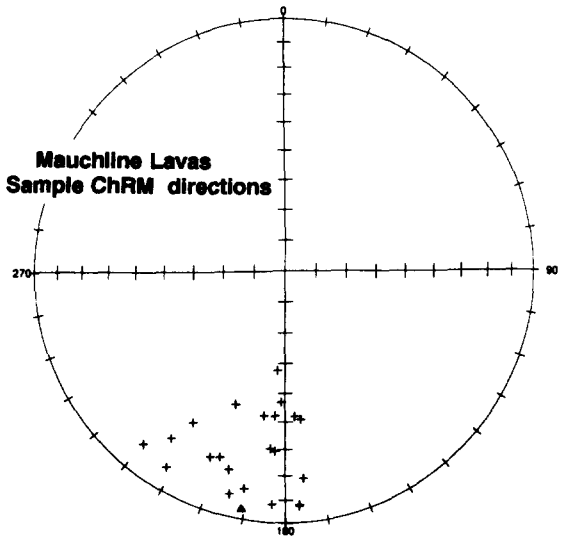


Fig. 5. In situ directions of high blocking temperature components of magnetisation in the Mauchline lavas. Equal area projection; crosses are upper hemisphere directions and the triangle is a lower hemisphere direction.

sessed. The lavas lie within a shallow depression (Fig. 1) which may, at least in part, record primary basement topography apparent today as the unconformity formed following deformation in the short interval between deposition of the Coal Measures and eruption of the Mauchline lavas. Thus it is not clear whether tilt adjustment is applicable to these lavas. The overlying red sediments have primary depositional dips and a component of small tectonic dip is not apparent. When adjusted for regional tilts ($< 10°$) implied by the outcrop pattern of the basin, grouping deteriorates slightly (Table 1) although the remanence direction and pole position are not significantly changed. Thus the fold test is not diagnostic and could suggest either that (i) remanence postdates deformation or (ii) deformation was essentially complete at the time of eruption.

Basic igneous rocks of Carboniferous and Permian age are widely distributed in the British Isles (Upton, 1982) and related to a succession of rifting episodes broadly synchronous with the Variscan Orogeny. Palaeomagnetic (south) poles derived from these rocks following progressive thermal demagnetisation and component analysis are summarised in Table 2. They include data from single dykes which are unlikely to have averaged secular variation. Furthermore, it is only the Lower Carboniferous units influenced by Variscan deformation which typically permit a palaeomagnetic fold test although contact tests are available for some of the dykes; hence the assumed primary or secondary origin of the remanence (Table 2) is not often established. However, a feature of the pole positions is a close grouping of Lower Carboniferous poles near 330°E, 15°S followed by a 20–30° southerly shift to a second group of Late Carboniferous–Permian data (Fig. 6) plotting near 340°E, 40°S. The Lower Carboniferous results show mixed polarities whilst the latter results are of uniform reversed polarity and are probably all derived from rocks magnetised during the PCRS. The Mauchline pole falls in this second group close to the poles from the Late Carboniferous Midland Valley Sill of central Scotland and the early Permian Exeter Lavas (280 Ma) in southern Britain.

3.3. Palaeointensity study

A modified version of the Thellier method, similar to that used by Prévot et al. (1985), was applied to 15 cores to derive estimates of ancient field intensity. The experimental procedure in-

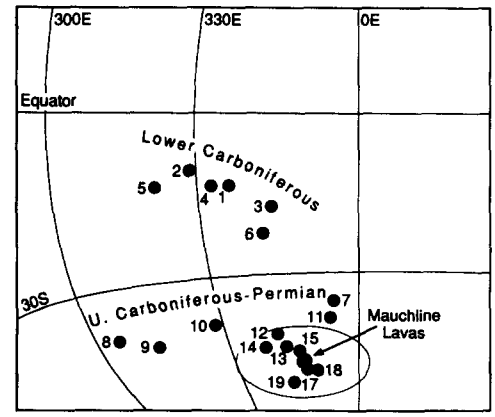


Fig. 6. Palaeomagnetic south poles derived from thermally cleaned Carboniferous and Permian rock units in the British Isles subject to component analysis. The poles are numbered according to Table 2. The 95% confidence oval is shown for the Mauchline result of this study.

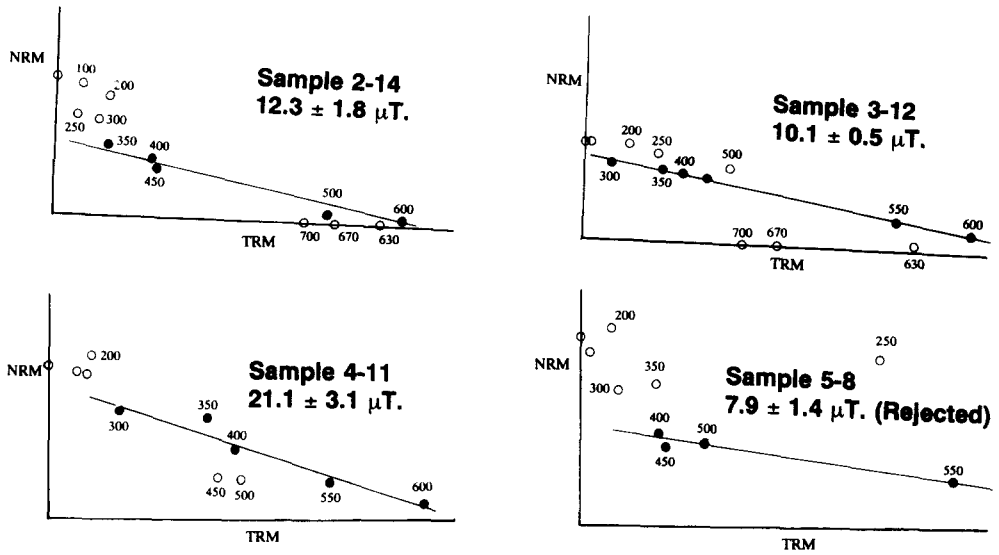


Fig. 7. Four examples of Arai plots from modified Thellier palaeointensity experiments. The NRM and PTRM values are normalised and the closed (open) circles define selected (rejected) points.

volves six steps: (i) the NRM is initially measured at room temperature; (ii) the sample is heated to a temperature T_1, maintained for 20 min before cooling in a zero field and the residual NRM is measured; (iii) the sample is reheated to T_1, in a controlled laboratory field of 50 μT, maintained for 20 min before cooling in the 50 μT field and the newly acquired PTRM is measured; (iv) step (ii) is repeated for $T_2(>T_1)$; (v) step (iii) is repeated for T_2; finally (vi) the sample is reheated to T_1 and cooled in a zero field and the remanence is measured to provide a partial thermore-

Table 3
Results of Thellier–Thellier palaeointensity experiments

Sample no.	Ba (μT)	% error	f	g	q	Accepted temperatures
1– 5 [a]	33.6 ± 15.7	46.7	0.412	0.346	1.31	350, 400, 450, 550
1– 6 [b]	129.9 ± 13.6	10.5	0.347	0.448	1.49	350, 400, 450
1–11 [a]	10.3 ± 3.1	31.1	0.558	0.413	0.75	350, 400, 450, 550
1–16 [b]	107.5 ± 6.9	6.4	0.371	0.480	2.78	350, 400, 450
2–13	8.6 ± 0.4	4.6	0.643	0.507	6.43	400, 450, 500, 550
2–14	12.3 ± 1.8	14.6	0.878	0.578	4.81	350, 400, 450, 500, 600
3–12	10.1 ± 0.5	5.0	0.697	0.655	10.07	300, 350, 400, 550, 600
3–15	17.3 ± 1.1	6.4	0.815	0.585	7.28	300, 350, 500, 550, 600
3–17	13.5 ± 1.0	7.4	0.804	0.671	7.13	300, 300, 450, 500, 550, 600
3–18	7.7 ± 0.6	7.8	0.522	0.421	1.76	300, 350, 400, 450, 500, 550
4– 9	35.5 ± 4.3	12.1	0.800	0.602	4.00	300, 350, 450, 550, 600
4–11	21.1 ± 3.1	14.7	0.674	0.715	3.25	300, 450, 500, 550, 600
4–17 [a,b]	10.7 ± 3.3	10.8	0.169	0.435	0.24	300, 350, 400
5– 8 [a]	7.9 ± 1.4	17.7	0.815	0.575	7.23	400, 450, 500, 550
5– 9	17.8 ± 1.1	6.2	0.721	0.343	3.91	400, 450, 500, 600

Samples failing selection criteria are indicated for reasons of insufficient points ([a]) or % error ([b]). f, g and q are the fraction, gap and quality factors, respectively, of Coe et al., (1978).

manent magnetisation (PTRM) check. Steps (ii) to (vi) are repeated to progressively higher temperatures until the Curie point is reached.

Results are analysed using Arai plots (Nagata et al., 1963) on which the remaining NRM at each stage of treatment is plotted against the PTRM acquired for that temperature step. The palaeointensity is determined from the gradient of the plot using a standard least squares fit of selected points; the error is determined from deviation of points from the best fit line.

To rate acceptance a determination was required to satisfy four criteria namely: (i) The blocking temperature spectrum must cover only a single convergent component of magnetisation as identified from study of orthogonal vector plots; in the present study this generally required exclusion of data below 300°C. (ii) The fraction of NRM, f, used in the calculation of the palaeointensity value must be $\geqslant 15\%$ of the NRM (Coe et al., 1978). (iii) A minimum of four points must form the basis of the best fit calculation. (iv) The standard deviation of the best fit line should be $< 15\%$.

Results of four typical experiments are illustrated in Fig. 7. Nine of 15 experiments satisfy the criteria (Table 3). Study of Site 1 (where remanence is totally dominated by hematite) yielded no acceptable determinations; the remaining four sites with magnetite and magnetite–Ti hematite assemblages yield a mean value of 16.0 ± 4.8 μT. Excluding the anomalously high result (35.5 μT) from Sample 4–9, a mean palaeointensity value of 13.6 ± 4.5 μT is obtained, with a corresponding virtual dipole moment (VDM) of 3.2×10^{22} Am^2. This is approximately 40% of the present field magnitude.

Fig. 7 shows that some samples exhibit a decrease in PTRM capacity above about 600°C. This behaviour is not typical of old lavas which tend to show an increase in PTRM capacity at high temperatures (e.g. Prévot et al., 1990; Perrin et al., 1991; Thomas, 1993) due to thermal alteration during laboratory heating. The low VDM value recorded by the Mauchline lavas receives some support from palaeointensity studies of other rocks formed during the PCRS (Rolph and Shaw, 1985; Oppenheim et al., 1994). Thomas et al. (1995a) report a VDM value of 3.3×10^{22} Am^2 for the Midland Valley Sill of central Scotland. Results from the Early Permian Caucasus in Russia indicate a great variation in field intensity over a short time period. Bolshakov et al. (1989) record VDM values of 7.2×10^{22} Am^2 and 3.8×10^{22} Am^2 for the beginning and end of the Early Permian respectively. Senanayake and McElhinny (1983) obtained a value of 6.2×10^{22} Am^2 from the 280 Ma Exeter traps of southern England and Senanayake (1981) quotes a value of 4×10^{22} Am^2 for the Late Permian Esterel volcanics. However, a comprehensive set of results for the PCRS field intensity from other Russian studies (Solodovnikov, 1992a,b) yield a VDM value close to, or slightly higher than, the present day value. The mean PCRS palaeointensity cannot be significantly modified by the mean of nine values for the Mauchline lavas reported here. However, the low VDM value obtained does add to the growing number of similar results being obtained for the PCRS (Thomas et al., 1995a,b). Thus, there appears to be a discrepancy in the Russian and other European results which we are, at present, unable to explain.

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